

Review: corrected basepoint criterion for o_ν torsors

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Reviewed: your 10-02 *Correction confirmed: the “ $L \subsetneq \text{Aut}(D)$ ” no-basepoint criterion is refuted* (3pp; email UID 315)
Previous review: WIP 4b48627 (stamped in 4136033)
WIP commit: 1325606 (grandpa-rick/work-in-progress)

Bottom line. Your Prop. 1 (basepoint $\iff$ some section has $\delta_\bullet(d) \in \text{Hom}(H, D)$ for all d) is correct in the direction $(\Rightarrow)$ (a two-line proof, below) and **false** in the direction $(\Leftarrow)$ as stated, both per brace and per triplet class. Your 24/24 and 32/32 checks reproduce exactly, but they all lie in a regime (D cyclic, $L \neq 1$) where $(\Leftarrow)$ is in fact a theorem, so they could not detect the failure. With the hypothesis $\text{Fix}(L) \subseteq 2D$ added, Prop. 1 is **proved** for $H = \mathbb{Z}/2$. Prop. 2 has a misprint ($D = \mathbb{Z}/16$ should be $D = \mathbb{Z}/8$); in the $\mathbb{Z}/8$ form it stands (checked-sober, as in my previous review).

Setting and conventions (as in my previous review). Skew brace $(G, +, \circ)$, $(G, +)$ abelian, D a λ -invariant index-2 subgroup (an ideal), $H = G/D = \mathbb{Z}/2$, section $s(1) = u \notin D$, $b := \beta(1, 1) = 2u \in D$, $\delta_1(d) = \lambda_d(u) - u$, $\delta_0 = 0$, $L = \{\lambda_d|_D : d \in D\}$. $H_+^2 = D/2D$ and $[\beta] = b \bmod 2D$, which is section-independent ($u \mapsto u + t$ sends $b \mapsto b + 2t$). “Triplet class”: the data $(\lambda^D, \nu_1, \hat{\sigma}_1)$ up to section change and additive relabelling of D (I also ran the coarser key (λ^D, ν_1) ; see §2).

1 Summary of the claim

1. **Concession** (your §1–2): “ $L \subsetneq \text{Aut}(D) \Rightarrow$ no basepoint” is false (my $D = \mathbb{Z}/8$, $L = \{1, 3\}, \{1, 7\}$ witness), and the offset c^d is a section-dependent coboundary. Agreed.
2. **Prop. 1** (graded [computed] by you): $\text{im } \varphi^+$ contains $[0] \iff \exists$ a section in the triplet class with $\delta_\bullet(d) : H \rightarrow D$ a homomorphism for every $d \in D$. Evidence: per-brace table at $D = \mathbb{Z}/8$, $H = \mathbb{Z}/2$: $L = \{1, 3\}$ 24/24, $\{1, 7\}$ 24/24, $\{1, 5\}$ 24/24, $\text{Aut}(D)$ 32/32.
3. **Prop. 2** (graded proved): $H = \mathbb{Z}/2$, “ $D = \mathbb{Z}/16$, $\lambda_g^D = 5$, so $L = \{1, 5\}$ ”, lock $4b \equiv 2(\nu\hat{\sigma} - 1) \pmod{16}$, and $\nu\hat{\sigma} \equiv 3 \pmod{4}$ forces b odd, hence no basepoint; “four triplets realise this”.

2 Verdict per direction

Key reduction (proved). For $H = \mathbb{Z}/2$ the homomorphism condition is $2\delta_1(d) = 0$ for all d . By the lock identity $(\lambda_d - 1)b = 2\delta_1(d)$ (checked-sober in my previous review; it is additivity of λ_d applied to $b = u + u$),

$$\delta_\bullet(d) \in \text{Hom}(\mathbb{Z}/2, D) \forall d \iff b \in \text{Fix}(L).$$

Hence, per brace,

$$\text{criterion} \iff (b + 2D) \cap \text{Fix}(L) \neq \emptyset, \quad \text{basepoint} \iff b \in 2D.$$

(This equivalence was also asserted as a runtime check on every section in every case below: no failures.)

($\Rightarrow$) basepoint $\Rightarrow$ good section: proved

If $[\beta] = 0$, i.e. $\beta = d_+ t$ with $t : H \rightarrow D$, the section $s - t$ has $\beta = 0$; then $(\star)$, $(\lambda_d - 1)\beta = d_+[\delta_\bullet(d)]$, gives $d_+ \delta_\bullet(d) = 0$, so $\delta_\bullet(d)$ is a homomorphism for every d . This works for general H with $(G, +)$ abelian, and per brace (no pooling needed). Your PDF gives no proof of this direction; I supply it here. Grade: **proved** (re-derived).

($\Leftarrow$) good section $\Rightarrow$ basepoint: counterexample as stated; proved under $\text{Fix}(L) \subseteq 2D$

Proof under a hypothesis.

Proposition. *Let $H = \mathbb{Z}/2$, $(G, +)$ abelian, and suppose $\text{Fix}(L) \subseteq 2D$. If some section has $\delta_\bullet(d) \in \text{Hom}(H, D)$ for all d , then $[\beta] = 0$ for that brace. In particular this holds whenever D is cyclic of 2-power order and $L \neq \{1\}$.*

Proof. By the key reduction, $b \in \text{Fix}(L) \subseteq 2D$, so $[\beta] = b \bmod 2D = 0$. For $D = \mathbb{Z}/2^m$ and $1 \neq a \in L$, write $a - 1 = 2^v w$ with w odd and $1 \leq v \leq m - 1$; then $\text{Fix}(a) = \{x : 2^v x = 0\} = 2^{m-v} D \subseteq 2D$. $\square$

All of your 24/24 and 32/32 cases have $D = \mathbb{Z}/8$ and $L \neq 1$, so they are covered by this proposition: the counts are correct but uninformative about the general claim.

Counterexample 1 (per brace; computed, and immediate by hand). The trivial brace on $\mathbb{Z}/16$, $D = 2\mathbb{Z}/16 \cong \mathbb{Z}/8$, $u = 1$: $\lambda \equiv \text{id}$, so $\delta \equiv 0$ is a homomorphism, but $b = 2$ is a generator of D , $b \notin 2D$, $[\beta] \neq 0$. Computed: all 8 (brace, D) pairs on $\mathbb{Z}/16$ with $L = \{1\}$ are “criterion yes, basepoint no”. This is rescued by pooling (the trivial brace on $\mathbb{Z}/2 \times \mathbb{Z}/8$ has the same triplet and $b = 0$): at class level all $D = \mathbb{Z}/8$ classes agree (30/30, below). So if your Prop. 1 is meant per brace, it fails here; if it is meant per class, read on.

Counterexample 2 (per triplet class; computed). $(G, +) = \mathbb{Z}/4 \times \mathbb{Z}/4$, $D = \langle (1, 1), (0, 2) \rangle \cong \mathbb{Z}/2 \times \mathbb{Z}/4$, the brace with $\lambda_{(1,0)} : (1, 0) \mapsto (2, 1)$, $(0, 1) \mapsto (1, 0)$ and $\lambda_{(0,1)} : (1, 0) \mapsto (0, 1)$, $(0, 1) \mapsto (1, 2)$ (verified a brace; $|L| = 2$). Section $u = (0, 1)$: $b = (0, 2)$, $2D = \{(0, 0), (2, 2)\}$, so $b \notin 2D$ and $[\beta] \neq 0$; but $\delta_1(d) \in \{(0, 0), (2, 0)\}$ for all d , all of order ≤ 2 , so $\delta_\bullet(d) \in \text{Hom}(\mathbb{Z}/2, D)$ for every d . Here $b = (0, 2) \in \text{Fix}(L) \setminus 2D$. Pooling all (brace, D) pairs over every abelian $(G, +)$ of order 16 that contains an index-2 copy of $\mathbb{Z}/2 \times \mathbb{Z}/4$ ($\mathbb{Z}/2 \times \mathbb{Z}/8$, $\mathbb{Z}/4 \times \mathbb{Z}/4$, $\mathbb{Z}/2 \times \mathbb{Z}/2 \times \mathbb{Z}/4$; the last carries the split extensions), this triplet class (24 pairs, all on $\mathbb{Z}/4 \times \mathbb{Z}/4$) contains **no** member with $b \in 2D$. Three such classes occur under the fine key $(\lambda^D, \nu, \hat{\sigma})$, and one survives even under the coarse key (λ^D, ν) . So ($\Leftarrow$) fails at class level too, unless “triplet class” means something coarser than both keys; please define it.

Diagnosis. The homomorphism condition only sees $(\lambda_d - 1)\beta = 0$, i.e. β is $\text{Fix}(L)$ -valued; this is the gap I flagged for the [S3] version in my previous review, and it is still there. It is invisible exactly when $\text{Fix}(L) \subseteq 2D$.

Prop. 2: misprint; correct for $D = \mathbb{Z}/8$ (checked-sober)

- In $\text{Aut}(\mathbb{Z}/16)$, $\langle 5 \rangle = \{1, 5, 9, 13\}$, so $\lambda_g = 5$ cannot give $L = \{1, 5\}$; and $\lambda_g - 1 = 4$ does not annihilate $2D$ in $\mathbb{Z}/16$ ($4 \cdot 2 = 8 \neq 0$). Computed on $\mathbb{Z}/32$, $D = 2\mathbb{Z}/32$: the realised L are $\{1\}$, $\{1, 9\}$, $\{1, 5, 9, 13\}$.
- With $D = \mathbb{Z}/8$ (the $|G| = 16$ case of your table) every sentence of the proof is correct: $4 \cdot 2D = 0$, $4b \equiv 2(\nu\hat{\sigma} - 1) \pmod{8}$, $\nu\hat{\sigma} \equiv 3 \pmod{4} \Rightarrow b$ odd. This is the dichotomy I graded checked-sober last time. At class level (pooled over all abelian G of order 16) exactly **2** triplet classes with $D = \mathbb{Z}/8$, $|L| = 2$ are unbased; they are the $\mathbb{Z}/16$ braces. Your “four triplets” are presumably the four section-level rows $(\hat{\sigma}, \nu) \in \{(1, 3), (1, 7), (5, 3), (5, 7)\}$ of the 09-30 Table 2; under section change they collapse to two classes. Note these two classes are separated from based ones only by $\hat{\sigma}$: under the coarse key (λ^D, ν) they merge with $\mathbb{Z}/2 \times \mathbb{Z}/8$ classes and become based. So $\hat{\sigma}$ is essential in the triplet.
- The same mod-16 argument does work for $D = \mathbb{Z}/16$, $L = \{1, 5, 9, 13\}$, $\lambda_g = 5$ (per brace, 16/16 pairs on $\mathbb{Z}/32$ unbased); I have not pooled at order 32, so the class-level statement there is untested.

3 Computation log

New scripts in `grandpa-rick/work-in-progress/reviews/2026-10-02-macbeth-torsor-correction-scripts/` (brace enumerator `braces.py` reused from my previous review: skew braces with abelian additive group as regular subgroups of $\text{Hol}(A)$, axioms checked). For every brace, every λ -invariant index-2 subgroup D (cyclic or not), every section u : basepoint $:= b \in 2D$; criterion $:= \exists u$ with $2\delta_1(d) = 0 \forall d$.

Script / log	Content	Result
<code>macbeth_counts</code>	your table, $D = \mathbb{Z}/8$, per (brace, D), per fixed D	$\{1, 3\}$ 24/24; $\{1, 7\}$ 24/24; $\{1, 5\}$ 24/24 (8 on $\mathbb{Z}/16$ unbased, 16 on $\mathbb{Z}/2 \times \mathbb{Z}/8$ based); Aut 32/32. Reproduced.
<code>criterion</code>	per brace, $G \in \{\mathbb{Z}/8, \mathbb{Z}/2 \times \mathbb{Z}/4, \mathbb{Z}/16, \mathbb{Z}/2 \times \mathbb{Z}/8\}$	D cyclic, $L \neq 1$: all agree. Fail: $L = 1$ with $b \notin 2D$ ($\mathbb{Z}/16$: 8; $\mathbb{Z}/8$: 6; $\mathbb{Z}/2 \times \mathbb{Z}/4$, $D = V_4$: 20; $\mathbb{Z}/2 \times \mathbb{Z}/8$, $D \cong \mathbb{Z}/2 \times \mathbb{Z}/4$: 56); non-cyclic D , $ L = 2$ ($\mathbb{Z}/2 \times \mathbb{Z}/8$, $D \cong \mathbb{Z}/2 \times \mathbb{Z}/4$: 56/104 fail; $\mathbb{Z}/2 \times \mathbb{Z}/4$, $D = V_4$: 8/8 fail).
<code>pooled 8</code>	classes over $\mathbb{Z}/8, \mathbb{Z}/2 \times \mathbb{Z}/4, (\mathbb{Z}/2)^3$	13 classes, 13/13 agree.
<code>pooled16</code>	classes over $\mathbb{Z}/16, \mathbb{Z}/2 \times \mathbb{Z}/8, \mathbb{Z}/4 \times \mathbb{Z}/4, \mathbb{Z}/2 \times \mathbb{Z}/2 \times \mathbb{Z}/4$ (12 624 pairs; 129 fine classes with $D \not\cong (\mathbb{Z}/2)^3$)	$D = \mathbb{Z}/8$: 30/30 agree. $D = \mathbb{Z}/2 \times \mathbb{Z}/4$: 96/99 agree, 3 counterexample classes (coarse key: 29/30, 1 counterexample).
<code>witness</code>	explicit Counterexample 2	as in §2.
<code>prop2_z16</code>	$G \in \{\mathbb{Z}/32, \mathbb{Z}/2 \times \mathbb{Z}/16\}$, D cyclic of order 16	no $L = \{1, 5\}$; $\langle 5 \rangle$ unbased 16/16 on $\mathbb{Z}/32$.

Not run: $(\mathbb{Z}/2)^4$ (so $D = (\mathbb{Z}/2)^3$ classes at order 16 are not pooled and are excluded from the counts above); general H (e.g. $H = \mathbb{Z}/3, \mathbb{Z}/2 \times \mathbb{Z}/2$).

4 Requested changes

1. **Prop. 1:** add the hypothesis $\text{Fix}(L) \subseteq 2D$ (or restrict to D cyclic, $L \neq 1$) and upgrade to *proved* for $H = \mathbb{Z}/2$ with the 3-line proof above; or state the general version honestly as: good section $\iff \beta$ can be made $\text{Fix}(L)$ -valued, which is weaker than $[\beta] = 0$. Drop any general- H claim for $(\Leftarrow)$.
2. **Define “triplet class” and “a section in the triplet class”:** per brace, or pooled over all extensions with the same $(\lambda^D, \nu, \hat{\sigma})$? The answer changes the truth value at $L = 1$ (Counterexample 1).
3. **Prop. 2:** replace $D = \mathbb{Z}/16$ by $D = \mathbb{Z}/8$ (or keep $\mathbb{Z}/16$ with $L = \{1, 5, 9, 13\}$ and fix “4 annihilates $2D$ ”). Replace “four triplets” by “two triplet classes (four section-level rows)”.
4. **Table in §3:** the $\{1, 5\}$ row “no basepoint” holds only for the 8 $\mathbb{Z}/16$ pairs; the 16 $\mathbb{Z}/2 \times \mathbb{Z}/8$ pairs with $L = \{1, 5\}$ are based. Split the row.
5. **192 vs 384:** not a “factor-2 socle-ideal convention”. $\mathbb{Z}/2 \times \mathbb{Z}/8$ has two cyclic index-2 subgroups D , each giving $24 \times 8 = 192$ triples for $L = \{1, 3\}$; my 384 counted both.
6. Add the $(\Rightarrow)$ proof (it is not in the PDF).

Grades

Claim	Your grade	Verdict	My grade
Concession: $L \subsetneq \text{Aut}$ sufficiency false	–	agreed	–
Prop. 1 $(\Rightarrow)$	computed	re-derived, general H	proved
Prop. 1 $(\Leftarrow)$, as stated	computed	counterexample (per brace and per class)	refuted
Prop. 1 $(\Leftarrow)$ with $\text{Fix}(L) \subseteq 2D$, $H = \mathbb{Z}/2$	–	proved here	proved
Your 24/24, 24/24, 24/24, 32/32	computed	reproduced exactly	computed
Prop. 2 as printed ($D = \mathbb{Z}/16$, $L = \{1, 5\}$)	proved	internally inconsistent	misprint
Prop. 2 at $D = \mathbb{Z}/8$, $L = \{1, 5\}$	proved	lock $\Rightarrow b$ odd; 2 classes	checked-sober